The Entangled Markets Hypothesis or The Butterfly Effect in a Normative, General Equilibrium for Asset Prices and Wages

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Abstract

The Efficient Markets Hypothesis specifies that asset prices reflect all available information, but that argument ignores the first step in portfolio management; namely, an asset re-allocation decision that was made to adapt to any changes in information. Investors also make deficient investment decisions for many reasons, and that, in turn, affects demand for the impacted assets (and potentially other assets). In a CAPM model, with so many free parameters (expected returns of the market portfolio, the risk-free asset, and volatilities), and with most variables treated as exogenous, the impact on one asset's price change is treated as largely unremarkable on another (except very tangentially through the market portfolio impact, if ever considered). CAPM and other asset pricing models largely ignore the fact that assets can be substitutes and complements of one-another (i.e., they are correlated); this is not true in practice and is also a potential shortcoming of tests of CAPM.

Instead, we make our point with the ("no free parameter") Goals and Risk-based Asset Pricing Model (GRAPM), which starts with optimal allocations to three classes of assets (absolute risk-free, relative risk-free and risky), to derive a normative general equilibrium model of asset prices and wages. Heterogeneity in this model is the result of each goals-based investor (e.g., retirement or education) desiring different cash flows. We perturb the allocation of one goals-based investor to their relative risk-free asset, and in a second case to the generic risky asset, (and by construct, the absolute risk-free asset). We demonstrate how it impacts, in a very non-intuitive way, the asset expected returns/prices, but also the profitability of firms and the wages they might offer. This simple, shelf-model variation of GRAPM opens the door to a rich vein of research given that investors make any number of Behaviorally Affected Decisions (BADs) that result in less-than optimal allocations to various assets or agents (external asset managers). We term this the Entangled Market Hypothesis; namely, we can have a GE outcome for asset prices and wages, different from "optimal" asset prices and wages, given deficient decision making by investors, but where the new equilibrium reflects how assets are entangled in a portfolio. We test this using Brazilian and US data, giving very different results/implications. Hence, EMH tests may need to go further to examine the spillover effects on assets other than just the one directly impacted by the event (e.g., others in the same sector; GIC)?

Keywords and JEL Classification

Keywords: Entangled Markets Hypothesis; Asset Pricing; Agency; Luck versus Skill; Goals and Risk-based Asset Pricing Model; Relative Asset Pricing Model; Stochastic Goals; Liability Driven Investing; Dual Normalization; Chameleons; Butterfly Effect; (No) Free Parameters; Heterogenous Investors; Janus Equilibrium; Goals-Based Investing; General Equilibrium;

JEL Classification: G1, G2, G12, C3, D2, D5, D14, J22, J3

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1. I received no financial support from any entities
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"Does the flap of a butterfly's wings in Brazil set off a tornado in Texas?

Edward Lorenz[1]

Entanglement is a ***quantum correlation*** *between two or more particles. If two particles are entangled, they can't be described independently, but only as a whole system.*[2]

Introduction

Nobel Laureate Eugene Fama's Efficient Markets Hypothesis or EMH (Fama 1965, 1970), argues that asset prices reflect all available information. EMH is based on a number of empirical tests that have withstood the test of time. The basis of all these tests is some variation of the Capital Asset Pricing Model (CAPM) of Sharpe (1964) or Merton (1973), because one needs an asset pricing model to test market efficiency. The empirical work on stock splits, earnings announcements, merger announcements, security issues, etc. "generally confirm that the adjustment of stock prices to events is quick and complete."[3] The tests were similarly applied to bond markets with similar results (Fama 2013). However, these tests typically left a few issues unexamined that we explore in this paper. For example, say the announcement for a stock split was positive for a stock, then one would expect that the commensurate rise in price observed in the empirical tests was typically because of an increase in demand (ruling out a reduction in supply). If an investor had a fully invested portfolio, then this increase in demand must come at the expense of some other asset for which there must be a reduction in demand. So, the driver of any asset price changes is an asset allocation decision, which EMH appears to not focus on. Further, shouldn't there be an impact on other asset prices, depending on their correlation to the primarily affected asset, because of a reallocation decision? If so, what are these impacts, because one would expect that there would be spillover effects for other assets that are substitutes (negatively correlated[4]), complements (positively correlated), or just liquidity pools (like cash or the risk-free asset)?

In this paper, we examine perturbations to optimal demand for assets in a normative general equilibrium (GE) model of assets and wages to highlight non-intuitive results that the EMH literature may have ignored, because as Fama (2013) notes, investors seem to be deficient in making effective investment decisions. "Even today, active managers (who propose to invest in undervalued securities) attract far more funds than passive managers (who buy market portfolios or whole segments of the market). This is puzzling, given the high fees of active managers and four decades of evidence that active management is a bad deal for investors."[5] Montier (2005) additionally catalogs the "Seven Sins of Fund Management" and provides further evidence of less-than-optimal decision making. But poor decisions on one asset or manager selected impacts a portfolio, which in turn must impact other assets as this decision works its way through the entire market of assets based on their respective correlations. We term this the Entangled Markets Hypothesis (EnMH)[6]; namely, we can have a GE outcome for all asset prices and wages, different from "optimal" asset prices and wages, given deficient decision making by investors because all assets appear to be "entangled" in reality. For example, on April 21, 2025, Zerohedge.com reported that, "In premarket trading, the Mag 7 are lower, with Tesla the top decliner: TSLA plunged 4% after Wedbush analyst and a Tesla bull Dan Ives warns of a "code red" moment ahead of first-quarter earnings (the rest are also hurting NVDA -3.1%, META -1.3%, AAPL -2%, AMZN -1.7%, GOOGL -1.3% and MSFT -0.9%)."[7] Hence maybe EMH tests need to go further and examine the spillover effects on assets other than just the one directly impacted by the event (e.g., others in the same sector; industry; GIC group)?

We propose a Pfleiderer (2020) "bookshelf model," to examine these issues, but we must abandon the focus on CAPM, despite its attractive features of providing practitioners with the three facets of investing: asset allocation, asset pricing, and risk-adjusted performance. We need a model with no "free parameters" (Cochrane 2005), and which reflects reality because CAPM ignores four realities of investing:

(i) that investors' goals are stochastic (and not a deterministic goal as assumed in Sharpe 1964, but rather as noted in Sharpe-Tint 1990, but sadly not reflected in the academic asset pricing literature as noted in Muralidhar 2025);

(ii) that investors have multiple stochastic goals like retirement, education, health etc., unlike single goal models in the literature (e.g., based on habit[8], peers/home bias[9], agency[10], background risk[11], or goals/liabilities[12]);

(iii) that investors (principals) typically seek to delegate to (hopefully) skillful agents (unlike the agency literature referenced earlier, which ignores skill of agents); and

(iv) they specify their objectives, especially their risk preferences, in a very focused/transparent manner, with clear absolute and relative risk goals (unlike the unobserved risk aversion parameter in current models).

The Goals and Risk-based Asset Pricing Model (GRAPM) of Muralidhar (2023, 2024b) is based on these four realities, satisfies the "no free parameter" solution (leading it to be termed a "Janus Equilibrium"), and is based on innovative new relative risk-free assets, launched by Brazil in 2023 (Merton and Muralidhar 2020, Merton, Muralidhar and Vitorino 2020a, 2020b).[13,14] In GRAPM, there are three classes of assets and can be of any asset class: (i) the absolute risk-free rate of CAPM (acronym, *F,* with zero volatility and correlation to all assets); (ii) the relative risk-free assets, that exactly mimic the stochastic goals of the investment pools for say retirement or a child's education (acronyms, *G1* and *G2*, respectively); and (iii) all other risky assets (generically, *I*, which could either be a single asset or a composite like a stock index or bond index). Since GRAPM starts with the optimal asset allocation of investors/consumers/workers to *F*, *G1*, *G2,* and *I* to derive equilibrium asset prices, we also know the wage that investors must earn to satisfy the budget constraint of purchasing their desired portfolio (i.e. the earnings they must have to buy the portfolio to meet their objective function). Because there are "no free parameters" in this model, asset prices and wages are "entangled", and hence a perturbation in demand for one asset (whether a relative risk-free asset or the generic risky asset, and by construct offset by the absolute risk-free asset) results in the entire set of equilibrium asset prices and wages changing – not just the single asset which experienced an "event".

Once we have an initial equilibrium of asset prices and wages, we can perturb this initial asset allocation assuming deficient behavior on the part of investors (what we can call Behaviorally Affected Decisions or BADs, discussed below and noted in Fama 2013 or Montier 2005), and examine how a change in a single asset allocation decision can impact the equilibrium price of F, $G1$, $G2$ and I, and hence the value of the portfolio and the earnings/wages of workers. This opens a rich new avenue for future research.

The objective function in GRAPM is taken from the actual Investment Policy Statements (IPSs) of real US institutional investors[15] (who represent the bulk of assets under management globally[16]), incorporating delegation to agents, and a desire for skillful agents. A critical issue for the GE model is that for say retirement investors, the relative safe asset for education is risky (as is the absolute risk-free asset, F, of the CAPM - Merton 2014), and for education investors, the relative risk-free asset for retirement (and F) is risky. This is simply because the cash flows for these goals are so different. Additionally, the relative risk-free retirement asset tailored to a 60-year-old is risky relative to the relative risk-free asset tailored to a 25-year-old (very easily understood by the reader given the duration mismatch of the cash flows). This is the "payoff perspective" (Cochrane 2022) at work, because each stochastic goal payoff is unique, and also ensures heterogeneity. For example, Brazil issued eight retirement relative risk-free assets (for investors aged 25, 30,…60), and 17 education relative risk-free assets (for children aged 18, 17, 16….etc.). Therefore, one can see how a change in asset allocation to say a relative risk-free for a 60-year-old, in principle, could have ripple effects through the market, even if miniscule, much like the butterfly flapping its wings in Brazil could impact weather in Texas.

This paper is different from GE models of say Merton (1973) that models an Intertemporal Capital Asset Pricing Model (ICAPM), but ignores the real economy and treats the investment opportunity set as stochastic. In reality, the goal for investing is the primary stochastic variable that drives investment decisions, as the burgeoning goals-based investing (GBI) literature has shown, and not the stochasticity of the opportunity set (and the basis for many of Fama (2013)'s justifications of their three and four factor models). Interestingly, Merton (1993) indirectly acknowledges the stochasticity of a goal in deriving investment decisions for a university endowment. Additionally, Merton (2014) reinforces the importance of stochastic goals in the context of defined contribution (DC) investing, arguing that it is not wealth that matters in DC plans, but retirement income! Retirement income is another phrase for matching a stochastic retirement liability cash flow. Lucas (1978), or even He and Krishnamurthy (2012), starts with an “endowment economy”, which includes an arbitrary production process and a perishable good, to derive asset prices, but this is an abstract model. Cox, Ingersoll and Ross (1985), CIR hereafter, who similarly model a real and financial economy, but the model is based on forecasts of real variables, which are notoriously difficult to identify or forecast. More recently, Gomes and Schmid (2017) offer a GE model “linking the pricing of stocks and corporate bonds to endogenous movements in corporate leverage and aggregate volatility.” Other GE models like Ai, Croce and Li (2013), with “investment options as intangible capital in a production economy”, Guasoni and Weber (2025), who model, “unhedgeable fundamental risk, combined with agents' heterogeneous preferences and wealth allocations,” and Zhang (2017)’s “Investment CAPM”[17], do not have similarities to our approach which starts with positive observations about reality to derive normative models of asset pricing and asset allocation (which many GE asset pricing models ignore). This GE model is also different from say demand-side models of Koijen and Yogo (2015) that provide “a portfolio choice model [that] implies characteristics-based demand when returns have a factor structure and expected returns and factor loadings depend on the assets’ own characteristics.” Our model is parsimonious and allows for simple perturbations as the asset prices are very clearly linked to the asset allocation decisions (and risk-adjusted performance calculations) – something missing from a bulk of the asset pricing literature.

The paper is structured as follows because we believe we achieve Cochrane (2022), when he states, “portfolio theory and application should describe a general equilibrium, and how investors are heterogeneous.” Section I highlights simple examples of BADs or Deficient Behavior as it pertains to the two key investment decisions: asset allocation and manager selection (which in turn will determine security selection which has the lowest impact in a portfolio). Section II then lays out the GE asset allocation, asset pricing and wage results of the GRAPM model using Brazil’s retirement bond for 25-year-olds (high duration), Brazil’s education bond for say 13-year-olds (lower duration), the S&P500 (in BRL terms), and the absolute risk-free asset. We use live data from 8/1/2023-11/29/2024 of Educa2030, RendA2060, and the S&P500 Total Return Index (in Brazilian Real terms), respectively, to represent these assets (along with a plug variable for *F*). We provide the key equations for GRAPM in Appendix I to keep the analysis/discussion strictly numerical. Section III demonstrates the impact of a small perturbation to the optimal allocation to a single relative risk-free asset on all these values and attempts to explain some of the counter-intuitive results (with equations in Appendix II). Section IV examines the less interesting case of perturbing the allocation to the generic risky asset (with equations in Appendix III). Section V suggests many possible extensions and new research avenues, and Section VI concludes.

I. Deficient Behavior By Investors

Having worked as an institutional asset owner (at a very sophisticated pension fund) and an asset manager to global institutional investors (at brand name firms and my own business) for over 30 years, the author has made his share of BADs and seen other investors do the same as well. Hence, Prof. Fama decrying an entire industry of active managers when passive management might have been cheaper and more effective is not a surprise. In fact, the firm that Prof. Fama is an advisor to (Dimensional Fund Advisors) also offers products that intelligent investors should probably not buy, so he is sensibly on the right side of that trade.

Montier (2005) identifies seven behavioral biases made by managers, including forecasting errors, trying to be experts at everything, meeting with firms, trying to "beat the gun", short horizons, accept stories as opposed to facts, and group decision-making. However, manager selection is the second tier of decision-making in most portfolios with delegated decision-making. The first and most critical decision is asset allocation, and that is where a lot of deficient decision-making originates, and where the bulk of the mis-allocation of risk emanates.

Asset Allocation. Arguably, the most critical decisions that investors make is the asset allocation decision (Brinson, Hood, Beebouwer, 1986), which accounts for over 90% of a portfolio's risk. Asset allocation decisions are often either made with flawed inputs or with an incorrect model. In the case of the former, the author was in-charge of an asset allocation exercise for a $10 billion global pension fund in 1998, and polled all the most sophisticated asset managers and asset-liability consultants about their 10-year forecasts for all assets. For simplicity, we will just focus on the equity expected return forecasts. To an institution, the forecast was 8-10% p.a. The 10-year realized return for US equity was negative! This validates a key concern highlighted by Markowitz (1952) that any such methods of creating optimal portfolios using sophisticated techniques can only be predicated if we have robust beliefs about expected returns, volatilities and correlations. Markowitz (1952) notes clearly that his model depends on "beliefs", but adds in footnote 7, "[T]his paper does not consider the difficult question of how investors do (or should) form their probability beliefs," and this lack of clarity and difficulty has plagued institutional and retail investing for decades.[18] GRAPM asset allocations are view-neutral in that the allocation to key assets as shown in Appendix I (equations 5 and 6) are independent of expected return forecasts.

In an example of incorrect use of asset allocation models, many defined benefit (DB) pension funds globally, despite having to meet a guaranteed retirement cash-flow and should be using a Sharp-Tint (1990) liability-driven investment (LDI) model, to set allocation. Instead, many of the named funds whose IPS we referenced earlier, typically only use an asset-only Markowitz (1952) mean-variance optimization (MVO), that ignores the liability cash flows and hence have massive asset-liability mismatches and risk (as evidenced in their overall low funded status despite discounting liabilities with an expected rate of return on assets rather than a liability-duration equivalent yield).[19] Merton (2014) similarly admonishes defined DC retirement investors from treating F as risk-free because as noted earlier, F is risky relative to one's desired retirement cash flows. Yet the US Department of Labor allows Target Date Funds (TDFs) or life-cycle funds to be the default investment alternative in DC pension plans (and given safe harbor from lawsuits), despite the fact that there is no real economic basis to the change in allocation. As Merton (2014) notes, to reallocate in a TDF based on a change in one's age, and a movement from one risky asset, equity, to another, bonds (mismatched in cash flows to the retiree's target cash flows) is a prescription for a retirement crisis! Hence, there is plenty of evidence of deficient behavior in asset allocation in institutional and retail portfolios.[20] But this has consequences for other assets in a portfolio context too.

Manager Selection. The manager hiring decision, namely hiring a delegated agent that you hope is skillful, is similarly fraught with error. First, and most common, asset owners and consultants chase last year's winners because they find it difficult to justify to their Boards why they should hire a manager who has underperformed for the last three years (i.e., the principal-agent problem). This is known as the "asset owner's curse" in the industry because managers, soon after being hired, start to underperform. By the time they are hired, the market cycle has changed and their style is out of favor. But the principal-agent challenges between Boards (often non-financial individuals) and investment staff and/or consultants fosters this behavior. In short, since we have no ability to forecast future performance of assets or managers, it is safer to recommend someone who did well in the past, and not get pilloried for backing a past losing track record.

Second, asset owners, author included (who worked for a large asset manager too), like to be wined and dined by managers despite the fact that the managers may not be the best-in-class. Hiring a large asset manager leads to invitations to fancy conferences in exotic locations, expensive dinners, sports events, etc., and potentially mediocre performance.[21] Third, the decision-makers may use incorrect metrics to make decisions. Many investors say that they choose managers based on Information Ratios (or Differential Sharpe – Sharpe 1994). Modigliani-Modigliani (1997) demonstrated how the Information Ratio can be easily gamed with zero effort or skill of managers, and Muralidhar (2001) demonstrated that none of the measures based on CAPM (Sharpe Ratio, Modigliani-Modigliani, or Graham-Harvey 1994, 1997) rank managers identical to measure based on skill (Seigel-Ambarish 1996).

And finally, the most recent trend has been to favor illiquid assets/managers because their portfolios are not marked-to-market (MTM). This nuance greatly reduces the quoted volatility of the portfolio, which then potentially overstates the return (especially in market downturns). It definitely impacts the Sharpe Ratio (Sharpe 1994), and understates the correlation of these assets to other risky, liquid assets. The lack of MTM is now seen as a feature and not a bug, and endowments have at least 40% in such assets (and definitely not adopting the Merton 1993 recommendation for endowment investing)[22], and public pension plans have allocated at least a third of their portfolios.[23] All these examples of BADs and deficient behavior are clearly observed and this impacts demand for not only these assets, but also others in competitive market for capital. It is this interaction that we seek to examine in the Entangled Markets Hypothesis (EnMH).

II. A Simple Numerical Equilibrium Simulation with Live Data

To show how this model works, we assume a relatively simple case of two goal replicating assets, *G1* (a retirement bond) and *G2* (an education bond), with given volatilities, and correlations, derived from actual data from Brazil from August 1, 2023 – November 29, 2024. Appendix I has all the equations and definitions of variables for this model, and the key point is that there are individual expected return formulae for *F*, *G1* and *G2*, and *I*, unlike CAPM that treats *F* and the market portfolio as exogenous. Moreover, there are no free parameters in this model also unlike CAPM. Muralidhar (2024) demonstrates a 3-goal case (and even the formulae for an N-goal case). However, the equilibrium conditions in Appendix I (especially A.12 and A.13) implies that the 3 correlations in the 2 goal-replicating asset case are "entangled" in that they cannot take any arbitrary value (and this will make it hard to test this normative model despite Brazil issuing such goal-replicating assets, until all investors adopt GRAPM).

While this is a normative model, we use real data to establish correlations and volatilities of RendA+ 2060 (starts paying in 2060 = *G1*) and Educa+2030 (starts paying in 2030 = *G2*). The risky asset is the S&P500 US Equity Index in Brazilian Real (or *I*). The key statistics on these assets is shown in Table 1. The historical (arithmetic) annualized returns, assuming a 260-day year, are also provided for completeness. We also provide the data on alternate Educa+ (2040) and RendA+ (2035) bonds as Brazil created eight (seventeen) of these retirement (education) bonds for various start and end dates of cash flows for all age groups. As of May 2025, over 500,000 investors had purchased these instruments voluntarily for a total AUM of R$6.5 bn, so there is real demand for these instruments.

The key inputs are highlighted in green. We assumed a 3% Target Tracking Error (or relative risk), and used actual values of σ(*G1*) = 35.98%, σ(*G2*) = 7.13% and σ(*I*) = 13.49%, where σ(*) represents the volatility of the asset. The relevant historical correlations ($\rho_{i,j}$) are Educa 2030-RendA 2060 (0.873), Educa2030-S&P500BRL (-0.142), and RendA2060-S&P500BRL (-0.162).

	Educa 2030	Educa 2040	RendA 2035	RendA 2060	S&P500 TR ($)	BRL	S&P500 (BRL)
Annualized Return	0.25%	-6.64%	-7.41%	-21.67%	22.45%	15.93%	38.38%
Annualized Risk	7.13%	14.45%	14.88%	35.98%	12.55%	11.30%	13.49%
Return/Risk	0.04	-0.46	-0.50	-0.60	1.79	1.41	2.85
	Correlation Matrix						
	Educa 2030	**Educa 2040**	**RendA 2035**	**RendA 2060**	**S&P500 TR ($)**	**BRL**	**S&P500 (BRL)**
Educa 2030	1.000	0.938	0.936	0.873	0.018	-0.190	-0.142
Educa 2040		1.000	0.988	0.923	0.007	-0.186	-0.149
RendA 2035			1.000	0.934	0.010	-0.188	-0.148
RendA 2060				1.000	0.000	-0.193	-0.162
S&P500 TR ($)					1.000	-0.362	0.626
BRL						1.000	0.500
S&P500 (BRL)							1.000

Table 1: Raw Data on Various Educa+ and RendA+ Bonds, S&P500 and the Brazilian Real (Aug 1, 2023 = Nov 29, 2024).

i. *The Asset Pricing Results and Comparison to GE Values*

We solve for r(*F*), E[r(*G1*)], E[r(*G2*)] and E[r(*I*)], and prices assuming a one-period world, as shown in Table 2a below. The data highlighted in yellow are the equilibrium expected (forward-looking) returns for each of the assets, given correlation and absolute and relative volatility assumptions. The bottom part of Table 2a also shows the implications of a fixed target tracking error and different volatilities on the target correlation for both goals, *G1* and *G2*. The price of each asset *j* = $\frac{1}{(1+E[rj])}$, and we also include the implied price of each asset in the last column (9), and highlighted in blue. Interestingly, while the model solves out, it also does not converge to a clean GE equilibrium solution unless we assume that the RendA2060-S&P500BRL correlation is -0.1417 as well, slightly different from reality. This is shown in the table with a dark orange highlight in the correlation section. This is because in an "entangled" model of correlations, historical correlations may not satisfy the equilibrium model conditions. Regardless, the returns from the model are very reasonable and not outlandish - the lower duration asset, Educa2030, has a lower expected return than RendA2060. Notice that this asset pricing model is different from the current asset pricing model offered by Tesouro Nacional that is used to price (and bid-ask spreads three times a day) these instruments daily based on an implied real yield curve.

	ASSET PRICING								
	Expected Return	Volatility	Return/ Risk	Return - r(F)	Correlation				Price
Asset					F	S&P500 TR	RENDA 2060	EDUCA 2030	
1	2	3	4		5	6	7	8	
F	2.02%	0			1	0	0	0	0.9802
S&P500 TR	8.29%	13.5%	0.61	6.27%	0	1	-0.1417	-0.142	0.9234
RENDA 2060	4.72%	36.0%	0.13	2.70%	0		1	0.873	0.9550
EDUCA 2030	2.55%	7.1%	0.36	0.54%	0			1	0.9751
Target Tracking Error		3%							
ρ(τ,G1)		0.997							
ρ(τ,G2)		0.911							

Table 2a. The 2 Goal Case with 3% Target Tracking Error, and solving for r(*F*), r(*RENDA 2060*), r(*EDUCA 2030*), and S&P500 in BRL terms.

Additionally, asset allocations to various assets for *G1* and *G2* investors are provided in Table 2b, and also the expected return of the risk-adjusted portfolios, generically, *A* (along with the excess over the respective relative risk-free assets, generically, *L*). We impose the wage/budget constraint for a GE model; namely, that these asset prices and wages must satisfy the production function for all three profit-seeking entities that issue *G1*, *G2* and *I*. As a result, we also report the total cost of both portfolios as that will also need to be the total wages paid by the entities that earn zero economic profit.

	RENDA 2060			EDUCA 2030
ASSET ALLOCATION				
	Allocation			Allocation
Asset			Asset	
F	-23.26%		F	-18.98%
S&P500 TR	22.41%		S&P500 TR	21.93%
RENDA 2060	100.84%		EDUCA 2030	97.04%
Total	100.00%		Total	100.00%
E[r(A)]	6.14%			3.91%
Portfolio Price or Wage for Investor	0.9420			0.9628
E[r(A)]-E[r(L)]	1.43%			1.36%

Table 2b: Asset Allocation for Both Investors and Total Cost of Both Portfolios

The allocations look somewhat similar to *I* for the two different goals, and the allocation to *F* is negative (i.e., leverage) for both goals as well. But that could be because the two goals are highly correlated, and the correlation of both goals to the risky asset is similar (the only real difference being the actual volatility). The high allocation to the relative risk-free asset follows because all risky assets relative to both goals have tracking errors that are much above the 3% target range (between 13%-30%). Tweaking the starting variables with different correlations and volatilities will be tested in future research.

The total funds raised by all four entities issuing securities (including *F,* that is treated as a government pass-through) are reported in Table 2c, as well as the likely wages paid to employees (absent a clearly specified production function). By assumption, the "wages paid" by the entity issuing instrument, *F*, E_F, is effectively the tax imposed on citizens to allow the entity to lend to investors to achieve their goals, but at a market rate of return.

	Asset Sale Value $	Total Assets Sold	Paid to Workers
F	-0.4140	-42.24%	-0.4140
S&P500 TR	0.4095	44.35%	0.4095
RENDA 2060	0.9630	100.84%	0.9630
EDUCA 2030	0.9463	97.04%	0.9463
	1.9048	2.0000	1.9048

Table 2c. Total Assets Sold, Value Raised by each Entity for Sale of Cash Flows based on Equilibrium Prices and Potential Wages Paid to Workers

Table 2c is interesting as in this simple case, the need for *G1* and *G2* investors to leverage implies that the government instead of issuing T-bills, has effectively injected liquidity into the market which is not unrealistic (e.g., the Federal Reserve Bank has introduced $6.9 trillion into the market as of December 2024 by buying up a lot of junk assets from weak financial institutions). The sum/net total of assets equals 200% and each entity has either a positive or negative inflow of cash to meet their financing needs for labor, capital and potentially economic profit.

III. A Simple Perturbation to the Allocation to One Relative Risk-free Asset

Assume for simplicity that individual S_{G2}, with goal *G2*, decides to allocate $-\varepsilon$ to their relative risk-free allocation (i.e., *G2*) and take that allocation from *F* (this can be seen in Appendix I, but adjusting the solution in equation A.6.a). We chose the simplest variation to show our results. In this case, we assumed that $-\varepsilon = +5\%$, so implicitly we are over-hedging our goal by an over allocation to *G2* by investor S_{G2}. This would be akin to a corporate pension fund increasing their liability hedge to dampen the income statement volatility impact of pension contributions. The equations from this perturbation are in Appendix II, and the results are as follows. We know from the previous discussion that *G2* has value as the goal-hedging asset for S_{G2}, but also potentially as a risky asset for the *G1* investor, S_{G1}, and vice-versa for *G1*. In reality, even *I* has some value in hedging liabilities as shown in Coqueret et al (2017), but as Table 1 shows, the correlation of *I* to *G1* and *G2* is negative (i.e., *I* is very risky), so this is unlikely to be the case using this specific Brazilian data. Despite this lack of value as a hedging asset, the asset allocation shown in Table 3a (which shows no real change except the impact of ε relative to Table 2b, and colored in dark red), has surprising and non-intuitive implications for expected returns and prices for all assets (as colored in purple), including for *I*, as shown in Table 3b. Because of the impact of a change in asset allocation, one can easily see that the risk-adjusted portfolio performance (*A*) is modified for both goals because of the impact on prices (and that also changes the cost of the portfolio) in the Entangled case.

ORIGINAL						ENTANGLED				
	RENDA 2060			EDUCA 2030			RENDA 2060			EDUCA 2030
ASSET ALLOCATION						ASSET ALLOCATION				
	Allocation			Allocation			Allocation			Allocation
Asset			Asset			Asset			Asset	
F	-23.26%		F	-18.98%		F	-23.26%		F	-23.98%
S&P500 TR	22.41%		S&P500 TR	21.93%		S&P500 TR	22.41%		S&P500 TR	21.93%
RENDA 2060	100.84%		EDUCA 2030	97.04%		RENDA 2060	100.84%		EDUCA 2030	102.04%
Total	100.00%		Total	100.00%		Total	100.00%		Total	100.00%
E[r(A)]	6.14%			3.91%		E[r(A)]	6.63%			3.17%
Portfolio Price or Wage for Investor	0.9420			0.9628		Portfolio Price or Wage for Investor	0.9375			0.9693
E[r(A)]-E[r(L)]	1.43%			1.36%		E[r(A)]-E[r(L)]	0.60%			0.57%

Table 3a: Asset Allocation for Both Investors and Total Cost of Both Portfolios

	ORIGINAL						ENTANGLED					DIFFERENCE	
	Expected Return	Volatility	Return/ Risk	Return - r(F)	Price		Expected Return	Volatility	Return/ Risk	Return - r(F)	Price	Yield Impact	Price Impact
Asset						Asset							
1	2	3	4	2'	9	1	2	3	4	2'	9	DIFFERENCE	
F	2.02%	0.00%			0.9802	F	1.78%	0.00%			0.9825	-0.24%	0.0023
S&P500 TR	8.29%	13.51%	0.61	6.27%	0.9234	S&P500 TR	4.31%	13.51%	0.32	2.52%	0.9587	-3.98%	0.0353
RENDA 2060	4.72%	35.98%	0.13	2.70%	0.9550	RENDA 2060	6.03%	35.98%	0.17	4.25%	0.9431	1.32%	-0.0119
EDUCA 2030	2.55%	7.13%	0.36	0.54%	0.9751	EDUCA 2030	2.60%	7.13%	0.36	0.82%	0.9746	0.05%	-0.0004

Table 3b. The 2 Goal Case with 3% Target Tracking Error and solving for r(*F*), r(*RENDA 2060*), r(*EDUCA 2030*), and S&P500 in BRL terms, with a 5% Over Allocation to r(*EDUCA 2030*).

To provide all the details before we summarize the key results, in Table 3c we provide the results for the entities issuing these securities and paying wages to individuals.

ORIGINAL				
	Asset Sale Value $	Total Assets Sold		Wages Paid to Workers
F	-0.4140	-0.4224		**-0.4140**
S&P500 TR	0.4095	0.4435		0.4095
RENDA 2060	0.9630	1.0084		0.9630
EDUCA 2030	0.9463	0.9704		0.9463
Total	1.9048	2.0000		1.9048

ENTANGLED				
	Asset Sale Value $	Total Assets Sold		Wages Paid to Workers
F	-0.4641	-0.4724		**-0.4641**
S&P500 TR	0.4252	0.4435		0.4252
RENDA 2060	0.9511	1.0084		0.9511
EDUCA 2030	0.9946	1.0204		0.9946
Total	1.9067	2.0000		1.9067

Table 3c. Total Assets Sold, Value Raised by each Entity for Sale of Cash Flows based on Equilibrium Prices, and Potential Wages Paid to Workers

There are five key results that stand-out from this analysis:

1. The increased allocation to *G2* by S_{G2}, led to the expected return of both *G1* and *G2* to increase, with r(*G1*) rising by more than r(*G2*) as shown in Table 3b. This seems counter-intuitive because ostensibly, an increased demand for *G2* should lead to a higher price for *G2* and hence a lower expected return. But this is the non-intuitive nature of this model as it incorporates the portfolio reallocation decisions of investors in all goals and not just *G2* in isolation (which the original tests of EMH focused on) and *G1* and *G2* are highly correlated (0.87) as shown in Table 1. So, the change in demand and price for one asset impacts the expected returns of potential substitutes – a feature not envisioned in CAPM.
2. The expected returns of *I* and *F* declined (Table 3b), with r(*I*) declining more than r(*F*). The impact on *F* is potentially understandable, but the reaction of *I* is not from the standard CAPM-type of lens. Since there is an increased allocation to *G2* as a hedge for S_{G2}, it has implications for S_{G1} for whom *G2* is a substitute risky asset to *I*. Hence,

I is impacted as well in this general equilibrium, and probably much more reflective of reality, again a facet not envisioned in CAPM.

3. The E[r(*A*)], or the risk-adjusted portfolio returns from combining the absolute risk-free, the relative risk-free and risky asset, declined for S_{G2}, but rose for S_{G1}. We can see this in Table 3a. However, this was not a cause of joy for S_{G1} because the excess over *L* declined in both as shown in the lower part of the table. One could argue that this is a sign of Pareto Optimality of the original solution because arguably risk-adjusted returns are lower for the same risk constraints by deviating from the original allocation? Notice that this is a general equilibrium solution – just a deficient one.
4. As a result, the cost of the *G1* portfolio declined (which combines asset allocation and price), but the cost of the *G2* portfolio rose, and in turn, the wages needed to support these portfolios changed. The total cost of both portfolios increased, which could be Pareto-improving if we can normalize wages per unit of work, and the unit of work is unchanged, but we do not have that formula in the model (as we do not have a clear work allocation function specified as cleanly by individuals as we have an investment objective in the published IPSs of institutional investors).
5. Finally, E_I, the entity issuing risky assets earns more (and pays more in wages assuming zero economic profit), without any change in quantity of issuance as shown in Table 3c. E_{G2} earns more from the increased demand for its assets; E_{G1} earns less despite having no change in issuance.

In short, a small flutter of a butterfly in Brazil has rippled through the entire economy and altered the General Equilibrium, asset prices and wages, and earnings of firms (as far away as the New York Stock Exchange in this simulation) – or a "Butterfly Effect". The one-time shock to an asset, as tested by the likes of Nobel Laureate Prof. Fama and others, are shown here, and potentially stabilized until a new shock, but there are shocks to other assets as well. We are not aware of tests of EMH that captured this "Butterfly Effect". It appears from purely observing the risk-adjusted performance of both goals, that both investors are less well-off in the Entangled case relative to the non-perturbed case ("original"), but we are unable to comment on their overall welfare as their wages earned has risen and we do not have a robust objective function for work (especially among the entities)-leisure decision.

IV. A Simple Perturbation to the Allocation to the Risky Asset by One Investor

The less interesting case is to assume that individual S_{G1}, with goal *G1*, decides to allocate $+\eta$ to their risky asset allocation (i.e., *I*) and take that allocation from *F* (this can be seen in Appendix I, but adjusting the solution in equation A.5.a). This could be an example of a fund over-allocating to illiquid assets. Similarly, Table 4a, 4b, and 4c compare the impact on asset allocations, asset prices and wages by assuming $+\eta = +5\%$.

	ORIGINAL			ENTANGLED			
	RENDA 2060		EDUCA 2030		RENDA 2060		EDUCA 2030
ASSET ALLOCATION				ASSET ALLOCATION			
Asset	Allocation	Asset	Allocation	Asset	Allocation	Asset	Allocation
F	-23.26%	F	-18.98%	F	-28.26%	F	-18.98%
S&P500 TR	22.41%	S&P500 TR	21.93%	S&P500 TR	27.41%	S&P500 TR	21.93%
RENDA 2060	100.84%	EDUCA 2030	97.04%	RENDA 2060	100.84%	EDUCA 2030	97.04%
Total	100.00%	Total	100.00%	Total	100.00%	Total	100.00%
E[r(A)]	6.14%		3.91%	E[r(A)]	4.80%		2.84%
Portfolio Price or Wage for Investor	0.9420		0.9628	Portfolio Price or Wage for Investor	0.9542		0.9724
E[r(A)]-E[r(L)]	1.43%		1.36%	E[r(A)]-E[r(L)]	0.04%		0.00%

Table 4a. Asset Allocation changes from assuming deficient behavior initiated through an increased allocation to *I* by investor S_{G1}.

	ORIGINAL						ENTANGLED					DIFFERENCE	
Asset	Expected Return	Volatility	Return/ Risk	Return - r(F)	Price	Asset	Expected Return	Volatility	Return/ Risk	Return - r(F)	Price	Yield Impact	Price Impact
1	2	3	4	2'	9	1	2	3	4	2'	9	DIFFERENCE	
F	2.02%	0.00%			0.9802	F	2.36%	0.00%			0.9770	0.34%	-0.0032
S&P500 TR	8.29%	13.51%	0.61	6.27%	0.9234	S&P500 TR	2.42%	13.51%	0.18	0.06%	0.9764	-5.87%	0.0529
RENDA 2060	4.72%	35.98%	0.13	2.70%	0.9550	RENDA 2060	4.76%	35.98%	0.13	2.41%	0.9545	0.05%	-0.0005
EDUCA 2030	2.55%	7.13%	0.36	0.54%	0.9751	EDUCA 2030	2.84%	7.13%	0.40	0.49%	0.9723	0.29%	-0.0027

Table 4b. Expected returns/asset price changes from assuming deficient behavior initiated through an increased allocation to *I* by investor S_{G1}.

ORIGINAL				ENTANGLED			
	Asset Sale Value $	Total Assets Sold	Wages Paid to Workers		Asset Sale Value $	Total Assets Sold	Wages Paid to Workers
F	-0.4140	-42.24%	**-0.4140**	F	-0.4615	-47.24%	**-0.4615**
S&P500 TR	0.4095	44.35%	0.4095	S&P500 TR	0.4818	49.35%	0.4818
RENDA 2060	0.9630	100.84%	0.9630	RENDA 2060	0.9626	100.84%	0.9626
EDUCA 2030	0.9463	97.04%	0.9463	EDUCA 2030	0.9436	97.04%	0.9436
Total	1.9048	2.0000	1.9048	Total	1.9265	2.0000	1.9265

Table 4c. Funds raised by asset issuing entities and wages paid changes from assuming deficient behavior initiated through an increased allocation to *I* by investor S_{G1}.

There are five key results that stand-out from this analysis:

1. The increased allocation to *I* by S_{G1}, led to the expected return of *F*, *G1* and *G2* to increase, with r(F) increasing the most as shown in Table 4b. The change in demand and price for *I* by one investor impacts the expected returns of potential substitutes (as G2 is a substitute for I for S_{G1}, and similarly for S_{G2}) – a feature not envisioned in CAPM.
2. The expected returns of *I* declined (Table 4b), which is intuitive given increased demand, but now we have a specific magnitude of change.
3. The E[r(*A*)], or the risk-adjusted portfolio returns from combining the absolute risk-free, the relative risk-free and risky asset, declined for both S_{G2} and S_{G1} (unlike the previous case). We can see this in Table 4a. Similarly, the excess over *L* declined in both as shown in the lower part of the table.
4. The cost of the *G1* and *G2* portfolios rose (which combines asset allocation and price), and in turn, the wages needed to support these portfolios changed.
5. Finally, E_I, the entity issuing risky assets earns more (and pays more in wages assuming zero economic profit), as shown in Table 4c. E_{G2} and E_{G1} earn less despite having no change in issuance.

V. Extensions

There are many extensions to this simple model. First, the correlations of *G1* and *G2* to *I* were similar and the two were highly correlated. We could pick a more diverse set of assets (see Appendix IV for a more US centric case), but we chose the Brazilian data as it was the first example we have of a country creating new relative risk-free assets (without having to construct such goal-replicating assets ourselves). Appendix IV shows how EnMH works differently in different markets based on goal structure and correlation profiles. Second, we conducted experiments, not reported here, with ε = +5%, +10% and 1%, and they offer interesting results to be reported in future research. Third, modifying *G1* by another perturbation will by and large be similar to the *G2* perturbation. Further, since the GRAPM model has been tested for 3 and N goals, one could conceivably test the model with the same perturbation, but with the 3-goal model to examine how these changes ripple through a broader market.

In addition, Muralidhar (2023) considers the following extensions:

(a) Different weights of investor types: Adding the weights of each type of investor in the economy.[24] In the simple model here, we assumed all investors had the same impact on the market and we know that this is not true. Adding in weights of different investor types is not difficult[25], just tedious;

(b) Multiple agents or multiple asset portfolios: Assume the agent's portfolio, *I*, is made up of multiple assets as opposed to a single risky (or alternatively, that an investor hires multiple agents to diversify the risk of being wrong). This just makes the model a bit more complicated. Because the correlation of a portfolio of assets to a goal-specific asset is nothing more than a weighted sum of each asset correlation to the goal-specific asset, the more complex model can be solved using M-cube risk-adjusted performance. Muralidhar (2001) demonstrates how this is achieved (in the context of hiring multiple agents), and further how assets that might have been considered valuable from a diversification perspective in an absolute return-risk world, may be sub-optimal in a relative risk world (and vice versa).[26] If *M* is the portfolio of risky assets, such that

$$r(M) = \sum_j w_j r(j), \text{ then } \rho(M,G1) = \frac{\sum_j w_j \rho(j,G1)\sigma(j)}{\sigma(M)}. \tag{1}$$

In this multi-asset (or multi-agent) setting, the allocation to each risky asset *j* that is in this portfolio $M = a{*}w_j$ and can be solved iteratively (where "a" is the allocation to the risky asset from the M-cube solution). This potentially leads to an interesting new research avenue as to how the optimal (multi-asset) risky portfolio in *G1* relates to the optimal risky portfolio in *G2*. This offers a new twist to the notion of diversification and also has some ability to explain why many pension funds experienced declines in funded status during the equity crises of 2000-2 and 2008. In short, these pensions used MVO (in an absolute return-risk space) to establish optimal portfolios (implicitly assuming as in Modern Portfolio Theory (MPT)/CAPM that the goal is deterministic and maximizing wealth), while ignoring the fact that they had a stochastic goal that essentially resembled a long-duration bond asset. Hence, assets that looked attractive from a diversification perspective in MPT (e.g., equities) proved to be highly risky and even negatively correlated to the goal during crises. These assets might not have been included in a GRAPM portfolio for a limited relative risk budget. Coqueret et al (2017) attempt to address this challenge of capturing the goal-hedging value of specific equities as opposed to holding a "market portfolio" proxy.

(c) Different Risk Budgets: it is possible to have different relative risk budgets for different goals as the original asset pricing equations are independent of $\rho(\tau, L)$, while the asset allocations are impacted by this variable. In the Entangled case, as shown in Appendix II, this variable shows up in the asset pricing equations.

(d) Multiple Possible Equilibria: While there are no free parameters, it is possible that there could be multiple equilibria for the given conditions (i.e., the equilibrium is not unique). This will be explored in future research, even without any perturbations.

V. Conclusion

We believe we have achieved the Cochrane (2022) challenge that "portfolio theory and application should describe a general equilibrium, and how investors are heterogeneous." GBI is slowly becoming the norm for investors, and investors have multiple stochastic goals, delegate to agents (that they hope are skillful), budget risk in a precise manner, and seek to maximize goal-relative risk adjusted returns. Countries like Brazil are creating unique instruments to match the cash flow of these goals, and these new securities have probably opened the door to additional financial innovation that is cash-flow focused. However, as shown earlier, investors are notorious for BADs and make deficient decisions.

We present a more realistic normative model, where investors invest to achieve multiple stochastic investment goals like retirement, saving for a child's education, health etc., and tweak the allocation of assets away from the optimal weights to reflect such deficient decisions and to test to see how it impacts other assets too. We test the implications from this normative GE model using live data from Brazil on these relative risk-free instruments and other generic risky assets, and inject a perturbance away from the optimal allocations. We found that with the live data, and given historical correlations in the US and Brazilian market, and a small perturbance in the goal-hedging activity of one investor, the implications ripple through the entire economy by way of revised expected returns/assets prices, not always in an intuitive manner, because assets can be simultaneously hedging assets for one goal and risky for another (i.e., substitutes and complements). This new equilibrium also creates revised finances for the entities issuing securities, and the wages paid by them to employees, again in potentially non-intuitive ways. This US example demonstrates how the EnMH works differently in different markets based on goal structure and correlation profiles. This approach offers new avenues for research as this approach is easily extended in many ways by relaxing some of the assumptions made to arrive at this model.

Adam Smith notes, "He is led by an invisible hand to promote an end which was no part his intention."[27] This paper shows how individuals maximizing their goal-specific, risk-adjusted return, and likelihood of hiring skillful agents, for a given risk budget and goal, but who make a small deviation away from an "optimal allocation", could have implications far beyond their imagination. Hence, we present the Entangled Markets Hypothesis for further development by others with hopefully new and interesting theoretical and practical implications.

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Appendix I – GRAPM Model

Figure A.1 captures the economy that we are modeling with two major actors. The primary actor is "individuals", who work for "entities" (the second) to earn a wage so that they can they achieve their stochastic investment/consumption goals. For simplicity, we assume that there are only two goals (say retirement and child's education), and two individuals/workers/consumers, say S_{G1} and S_{G2} (and they can delegate the investments to each other for no fee for simplicity and it doesn't prevent them from working for the entities). We could relax this assumption later without loss of generality but it keeps the model simple.

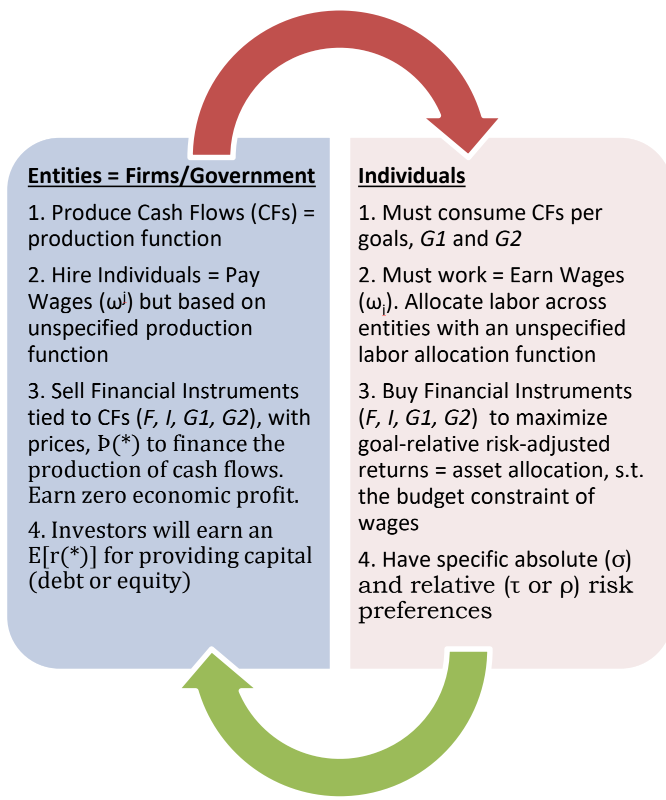

Figure A.1: The Circular Economy of Production and Consumption of Cashflows

The "commodity" that they consume is the unique set of cash flows (CF) needed to achieve their goals. These specific CFs are produced by two entities (say E_{G1} and E_{G2}), but there are also two more entities: one, say a government, that issues *F* (the absolute risk-free asset) and potentially with a welfare maximizing rather than profit motive; the other produces a risky cash flow, I, that is risky for both goals. So, this model can be seen as a GE financing model with entities needing to finance their "production" activities (with an unspecified production function), and individuals needing to finance their desired consumption (with a clear objective function, but unspecified utility and labor allocation functions). Despite this lack of specification of the various functions central to traditional economic and finance theory, we get very specific prices for securities and conditions for equilibrium wages.

All these key parameters can be summed up in Table A.1.

CRITERIA	E_{G1}	E_{G2}	E_I	E_F	S_{G1}	S_{G2}
Maximize	Profit	Profit	Profit	Welfare	Utility	Utility
Earnings/ Receipts	Þ$(G1)$*Amount Sold	Þ$(G2)$*Amount Sold	Þ(I)*Amount Sold	Þ(F)*Amount Sold and Taxes	Wages (ω_{G1}) & Cash Transfers from E_F	Wages (ω_{G2}) & Cash Transfers from E_F
Payments/ Outflows	Wages (ω^{G1})	Wages (ω^{G2})	Wages (ω^{I})	Þ(F)*Amount Bought and Cash Transfers	Value of the Portfolio (& any taxes to fund govt lending)	Value of the Portfolio (and any taxes)
Primary Goal Assumption	Zero Economic Profit	Zero Economic Profit	Zero Economic Profit	Social Welfare	Maximize Relative to Goal Return	Maximize Relative to Goal Return
Solve For:	Þ$(G1)$	Þ$(G2)$	Þ(I)	Þ(F)	Asset Prices, allocation and total wages (ω_{G1})	Asset Prices, allocation and total wages (ω_{G1})

Table A.1: Summary Parameters and Criteria in the GE model

Following the IPS of investors like LACERA or NMPERA, investors maximize expected, goal-relative, (net) risk-adjusted returns. Mathematically,

$$\text{Max E}[\text{r}(A) - \text{r}(L)], \tag{A.1}$$

where the risk-adjusted portfolio, A, is composed of the agent's portfolio, P, the goal hedging portfolio (or relative risk-free asset), L (to represent a generic version of G1 and G2), and F (absolute risk-free asset) as in equation (A.2), subject to equations (A.3) and (A.4). Equation (A.1) is Sharpe-Tint (1990) objective function, equations (A.2) and (A.3) are a slightly modified version of Modigliani-Modigliani (1997) and equation (A.4) is from Muralidhar (2001) to reflect the fact that investors (should) state very clearly, or worry about implicitly, tracking error budgets.

$$E[r(A)_{P,\,L|\tau}] = a^{P}_{\tau/L} \times E[r(P)] + l^{L}_{\tau/L} \times E[r(L)] + (1 - a^{P}_{\tau/L} - l^{L}_{\tau/L}) \times r(F) \tag{A.2}$$

$$\text{s.t. } \sigma(A) = \sigma(L) \tag{A.3}$$

$$\text{and } TE(A, L) = TE(\text{Target}, L) \text{ or } \tau \tag{A.4}$$

where $a^{P}_{\tau/L}$ is the allocation to the risky (agent) portfolio, P, given the goal of L, and a target relative risk of τ; $l^{L}_{\tau/L}$ is the allocation to the goal-hedging/goal-replicating asset, L (and measures what is *de-facto* invested in the low-cost passive benchmark or "beta" in industry parlance). The super-script denotes the asset being allocated to; the subscript indicates the target relative risk (τ) and the goal-replicating asset (L). Similarly, $E[r(A)_{x,\,y|\tau}]$ is the expected return of A, assuming x is the risky asset, y is the goal, for τ target risk. The balance of the assets is invested in the traditional absolute risk-free asset, F (and measures leverage too).

The asset allocation results are as follows:

$$a^{P}_{\tau/L} = \frac{\sigma(L)}{\sigma(P)}\left[\sqrt{\frac{[1-\rho(\tau,L)^2]}{[1-\rho(P,L)^2]}}\right] = \frac{\sigma(L)}{\sigma(P)}\left[\frac{\varphi(\tau,L)}{\varphi(P,L)}\right], \text{ where } \varphi(I,J) = \sqrt{[1-\rho(I,J)^2]} \tag{A.5a}$$

$$l^{L}_{\tau/L} = \rho(\tau,L) - \rho(P,L) \times \left[\sqrt{\frac{[1-\rho(\tau,L)^2]}{[1-\rho(P,L)^2]}}\right] = \rho(\tau,L) - \rho(P,L) \times \left[\frac{\varphi(\tau,L)}{\varphi(P,L)}\right] \tag{A,6a}$$

Which then gives the allocation to F. In industry parlance, l measures the beta (or benchmark allocation implicit in the portfolio, which is costlessly achieved), and (1-a-l) measures leverage (again, which is costless and should not be compensated). The optimal allocations of "a" and "l" as derived in equations (A.5a) and (A.6a) can be re-written in vector space (Muralidhar 2024c) as in equations (A.5b) and (A.6b), respectively.

$$a^{P}_{\tau/L} = \frac{\sigma(L)}{\sigma(P)}\left[\sqrt{\frac{[1-\rho(\tau,L)^2]}{[1-\rho(P,L)^2]}}\right] = \frac{\sigma(L)}{\sigma(P)}\left[\frac{sin\theta_{\tau,L}}{sin\theta_{P,L}}\right] \quad \text{(A.5b)}$$

And

$$l^{L}_{\tau/L} = \rho(\tau,L) - \rho(P,L) \times \left[\sqrt{\frac{[1-\rho(\tau,L)^2]}{[1-\rho(P,L)^2]}}\right] = cos\theta_{\tau,L} - cos\theta_{P,L} \times \left[\frac{sin\theta_{\tau,L}}{sin\theta_{P,L}}\right] \quad \text{(A.6b)}$$

If $\rho(P,L)$ is the correlation of returns of L and P of each agent (which is easily calculated from actual data), and $\rho(\tau,L)$ is the target correlation[28], which is specified by the principal, we can solve for the optimal asset allocation "*a*" as in equation (A.5a), and "*l*" in equation (A.6a). From (A.7a), the allocation to risky assets is an increasing, linear function of the volatility of the goal, a decreasing nonlinear function of the target correlation (i.e., risk aversion), and increasing, nonlinear function to the correlation of the asset to the goal. Conversely, it is a decreasing function of its own volatility. In short, *a* and *l* represent demand functions that we can perturb to run our ENMH simulations.

Note that in the M-cube approach, the investor decides which asset(s) form the risky portfolio. This freedom to choose what is "risky" is used to derive GRAPM and reflects reality. For example, one fund may decide that private credit is the risky asset they want to allocate to, and the second fund may decide that private equity is the risky asset of choice as there is no unique "market portfolio" in practice as established/required in CAPM.

Similarly, from (A.6), the allocation to the goal hedging asset is related positively and non-linearly to the target correlation, and negatively and non-linearly to the correlation of the risky asset to the goal. The allocation to the goal-replicating asset is independent of any volatility terms. These results are intuitively obvious as the allocation to risky assets should increase with increases in absolute and relative risk targets. Similarly, the allocation to the goal hedging asset will increase, the lower the relative risk target.

Interestingly, "*a*" and "*l*" will feature in the asset pricing equation as a way of capturing the dual value of an asset in serving as a risky asset for some goals, and hedging some part of other goals, respectively. And this demonstrates the power of GRAPM and the results of the ENMH model.

We can create two risk-adjusted portfolios relative to the *G1* goal, both with identical volatility and correlation to *G1 (*say Brazil's RendA+ retirement bond); one using *G2* (say Brazil's Educa+ education bond) as the risky asset and the other using *I* (say a single stock or the entire index like the S&P500 Equity Index in Brazilian Real terms) as the risky asset. While *I* and *G2* are unique, their risk-adjusted returns, created by a linear combination of that asset, *G1* and *F*, must be identical otherwise there would be zero demand for the asset with the lower risk-adjusted return (and those assets would cease to exist leaving only assets that satisfy this condition). Repeating this procedure for both S_{G1} and S_{G2} gives the equilibrium.

If we define

$$X_{I,G1,G2} = \varphi(I,G1) - \rho(I,G2) \times \varphi(I,G1) + \rho(G1,G2) \times \varphi(I,G2) \quad \text{(A7)}$$

$$Y_{I,G1,G2} = \varphi(I,G2) - \rho(I,G1) \times \varphi(I,G2) + \rho(G1,G2) \times \varphi(I,G1) \quad \text{(A.8)}$$

Then for the absolute risk-free asset,

$$r(F)_{G1,G2} = \frac{\left\{E[r(G1) - \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}}) \times E[r(G2)]\right\}}{\left[1 - \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}})\right]} \quad \text{(A.9)}$$

Note that $r(F)_{G2,G1}$ while different in terms of formula, must give the same value (and we show this in the numerical example below using actual Brazilian data).

For the goal-replicating assets, equation (A.10) must hold,

$$E[r(G1) - r(F)] = \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X_{I,G1,G2}}{Y_{I,G1,G2}}) \times E[r(G2) - r(F)] \quad \text{(A.10)}$$

And finally, for all risk assets, the following equation would apply:

$$E[r(I) - r(F)] = Z_{I,G1/G2} \times E[r(G1) - r(F)] \quad \text{(A.11)}$$

Where Zeta is a form of a "covariance term" as beta is in CAPM, defined as

$$Z_{I,G1/G2} = \frac{\sigma(I)}{\sigma(G1)} \times \left\{\left[\frac{\varphi(I,G2)}{\varphi(G1,G2)}\right] + (\frac{Y_{I,G1,G2}}{X_{I,G1,G2}}) \times \rho(I,G2) - (\frac{Y_{I,G1,G2}}{X_{I,G1,G2}}) \times \rho(G1,G2) \times \left[\frac{\varphi(I,G2)}{\varphi(G1,G2)}\right]\right\} \quad \text{(A.12a)}$$

Notice, that Zeta is a function of two volatilities, $\sigma(I)$ and $\sigma(G1)$, (as in CAPM), but three correlations: $\rho(G1, G2)$, $\rho(I, G2)$ and $\rho(I, G1)$. In CAPM, there is just a single correlation because CAPM is a highly specific/restrictive model that ignores the realities and relative decision-making of practice. One other key point to note, equation (A.11) cannot be used for assets that are goal-replicating; instead, equation (A.10) must be used for goal-replicating assets.

Interestingly, Zeta can be restated in terms of "a" and "l" as follows and has been interpreted in Muralidhar (2023) to show that the value of an asset is its role as a goal hedging asset, but also potentially as a risky asset for other goals.

$$Z_{I,G1/G2} = \frac{a^{G1}_{\tau/G2}}{a^{I}_{\tau/G2}} + \frac{\sigma(I)}{\sigma(G1)} \times \left(\frac{Y_{I,G1,G2}}{X_{I,G1,G2}}\right) \times l^{G2}_{I/G2} \quad \text{(A.12b)}$$

Similarly, express E[r(I)] as a function of E[r($G2$)] as the other risky asset, r(F), volatilities and correlations (assuming $G1$ is the goal) as in equation (A.13).

$$E[r(I) - r(F)] = Z_{I,G2/G1} \times E[r(G2) - r(F)] \quad \text{(A.13)}$$

We can see these dual equations in (A.11) and (A.13) for I as a pair-wise or "Janus" equilibrium.[29]

Appendix II – Impact on Equations from a Perturbation ε to Allocation to G2 by S_{G2}

We begin by perturbing just the G2 allocation for S_{G2}

$$l^{G2}_{\frac{\tau}{G2}} = \rho(\tau, G2) - \rho(P, G2) \times \left[\sqrt{\frac{[1-\rho(\tau,G2)^2]}{[1-\rho(P,2)^2]}}\right] - \boldsymbol{\varepsilon}$$

$$= \rho(\tau, G2) - \rho(P, G2) \times \left[\frac{\varphi(\tau,G2)}{\varphi(P,G2)}\right] - \boldsymbol{\varepsilon} \qquad \text{(A.14)}$$

Keeping earlier definitions of $X_{I,G1,G2}$ and $Y_{I,G1,G2}$, define $X'_{I,G1,G2}$ and $Y'_{I,G1,G2}$ that incorporate ε and $\varphi(\tau, G2)$ because the allocation, l to $G2$ was perturbed.

$$X'_{I,G1,G2} = \varphi(I,G1)^* \varphi(\tau,G2) - \rho(I,G2) \times \varphi(I,G1)^* \varphi(\tau,G2) + \rho(G1,G2) \times \varphi(I,G2)^* \varphi(\tau,G2)$$

$$- \quad \varepsilon^* \varphi(I,G2)^* \varphi(I,G2) \qquad \text{(A.15)}$$

$$Y'_{I,G1,G2} = Y'_{I,G1,G2} / \varphi(\tau,G2) \qquad \text{(A.16)}$$

Then

$$r(F)_{G1,G2} = \frac{\left\{E[r(G1) - \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X'_{I,G1,G2}}{Y'_{I,G1,G2}}) \times E[r(G2)]\right\}}{\left[1 - \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X'_{I,G1,G2}}{Y'_{I,G1,G2}})\right]} \qquad \text{(A.17)}$$

And for the relative risk-free assets

$$E[r(G1) - r(F)] = \frac{\sigma(G1)}{\sigma(G2)} \times (\frac{X'_{I,G1,G2}}{Y'_{I,G1,G2}}) \times E[r(G2) - r(F)] \qquad \text{(A.18)}$$

And for risky asset I, we created a new equilibrium equation

$$E[r(I) - r(F)] = Z'_{I,G2/G1} \times E[r(G2) - r(F)] \qquad \text{(A.19)}$$

Where

$$Z'_{I,G2/G1} = \frac{\sigma(G1)}{\sigma(G2)} \times \frac{1}{\varphi(G1,G2)} \times \{ \; [\varphi(\tau, G1)] \; + \; (\rho(\tau, G1) * \varphi(G1, G2) - \rho(G1, G2) \times \varphi(\tau, G1))(\frac{X'_{I,G1,G2}}{Y'_{I,G1,G2}}) \; \} \qquad \text{(A.20)}$$

Notice, that Zeta prime is a function of two volatilities, but now target correlations have entered the equation: $\rho(\tau, G2)$, and $\rho(\tau, G1)$ along with ε.

Appendix III – Impact on Equations from a Perturbation η to Allocation to I by S_{G1}

We begin by perturbing just the I allocation for S_{G1}

$$a_{\tau/L}^{P\eta} = \frac{\sigma(L)}{\sigma(P)}\left[\sqrt{\frac{[1-\rho(\tau,L)^2]}{[1-\rho(P,L)^2]}}\right] + \eta = \frac{\sigma(L)}{\sigma(P)}\left[\frac{\varphi(\tau,L)}{\varphi(P,L)}\right] + \eta \qquad \text{(A.21)}$$

Define $K = \frac{a_{G1}^{G2} * a_{G1}^{I\eta} - a_{G2}^{I}(l_{G2}^{G1} - l_{I\eta}^{G1})}{a_{G1}^{G2} * a_{G2}^{I} - a_{G1}^{I\eta}(l_{G1}^{G2} - l_{I}^{G2})}$

Then

$$r(F)_{G1,G2} = \frac{\{E[r(G2) - K \times E[r(G1)]\}}{[1-K]} \qquad \text{(A.22)}$$

And for the relative risk-free assets

$$E[r(G2) - r(F)] = K \times E[r(G1) - r(F)] \qquad \text{(A.23)}$$

And for risky asset I, we created a new equilibrium equation

$$E[r(I) - r(F)] = \left[\frac{l_{G2}^{G1} - l_{G2}^{I}}{a_{G2}^{I}} * K + \frac{a_{G2}^{G1}}{a_{G2}^{I}}\right] \times E[r(G1) - r(F)] \qquad \text{(A.24)}$$

Appendix IV – A US Centric Example

Muralidhar and Shin (2013) considers the case of US pension plans with say Risk Parity as the risky asset (because it is a multi-asset product), *I*, a thirty-year bond to represent the relative risk-free asset/liability of a typical corporate pension fund with a duration of about 15 years (*G1*) and a typical 60/40 portfolio to represent the SAA of a typical public pension plan (G2). Using those input values and a target tracking error of 3.5% p.a., we get results that are not too dissimilar from those of typical US pension funds.

The results are reported below in Table A.IV.1 and by comparison to the Brazil case, r(F) is lower, but the expected returns of the three other assets are higher. But this is because *G1* and *G2* are pretty much uncorrelated, and *I* is reasonably correlated to both liability streams.

	ASSET PRICING									
	Expected Return	Volatility	Return/ Risk	Return - r(F)	Correlation				Price	
Asset					F	RISK PARITY	LIABILITIES	60/40		
1	*2*	*3*	*4*	*2'*	*5*	*6*	*7*	*8*	*9*	
F	1.80%	0			1	0	0	0	0.9823	
RISK PARITY	9.60%	10.0%	0.96	7.79%	0	1	0.31	0.59	0.9124	
LIABILITIES	5.44%	6.0%	0.91	3.64%	0		1	-0.09	0.9484	DURATION 15
60/40	5.28%	8.0%	0.66	3.47%	0			1.00	0.9499	DURATION 5 ON BONDS
Target Tracking Error		3.5%								

Table A.IV.l: Analysis Using US Pension Fund Data

The perturbations of epsilon and nu also give different results: (i) a 5% over-allocation to the 30-year bond (which ripples more through Risk Parity than the traditional 60/40 portfolio) leads to a lower expected return of ALL assets and leads to r(A) underperforming both goals; and (ii) an equivalent over-allocation to nu of 5% (to Risk Parity) by S_{G1} leads to a near equalization of ALL asset returns (including F), and bare achievement of risk-adjusted performance relative to BOTH goals. This is a simple example of how the EnMH works differently in different markets based on goal structure and correlation profiles.

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[25] If w_{G1} is the weight of *G1* and w_{G2} is the weight of *G2* (such that the sum is 1), $X_{I,G1,G2}^{w_{G1},w_{G2}} = w_{G1} \times \varphi(I,G1) - w_{G2} \times \rho(I,G2) \times \varphi(I,G1) + w_{G2} \times \rho(G1,G2) \times \varphi(I,G2)$ and $Y_{I,G1,G2}^{w_{G1},w_{G2}} = w_{G2} \times \varphi(I,G2) - w_{G1} \times \rho(I,G1) \times \varphi(I,G2) + w_{G1} \times \rho(G1,G2) \times \varphi(I,G1)$.
[26] Muralidhar (2001) goes further to demonstrate how this approach is superior to naïve maximization of differential Sharpe ratios (also known as Information ratios) and other techniques that do not rank multiple agent portfolios consistent with measures of confidence in skill.
[27] https://www.panmurehouse.org/adam-smith/smith-quotes-faqs/#:~:text=quotes%20from%20Smith-,'No%20society%20can%20surely%20be%20flourishing%20and%20happy%2C%20of%20which,'
[28] Which is an exogenous relative risk target value specified by the principal and derived from TE(Target) once $\sigma(L)$ is known.
[29] In this example, the Janus duality is demonstrated by the fact that *I* can be priced in this pair-wise equilibrium with *G1* as risky (*G2* as goal) or vice versa. This duality is unchanged even when we add more goal-replicating assets, as we only use two at a time to create the asset pricing equation for *I*.